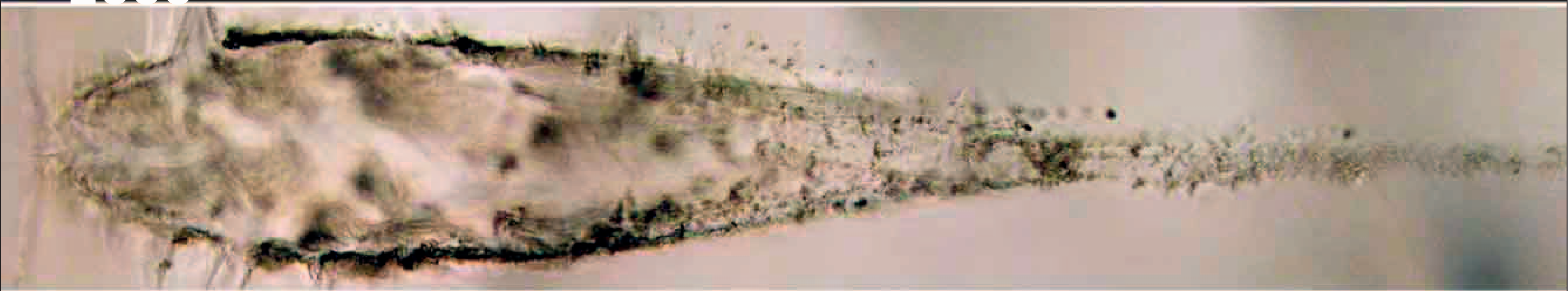
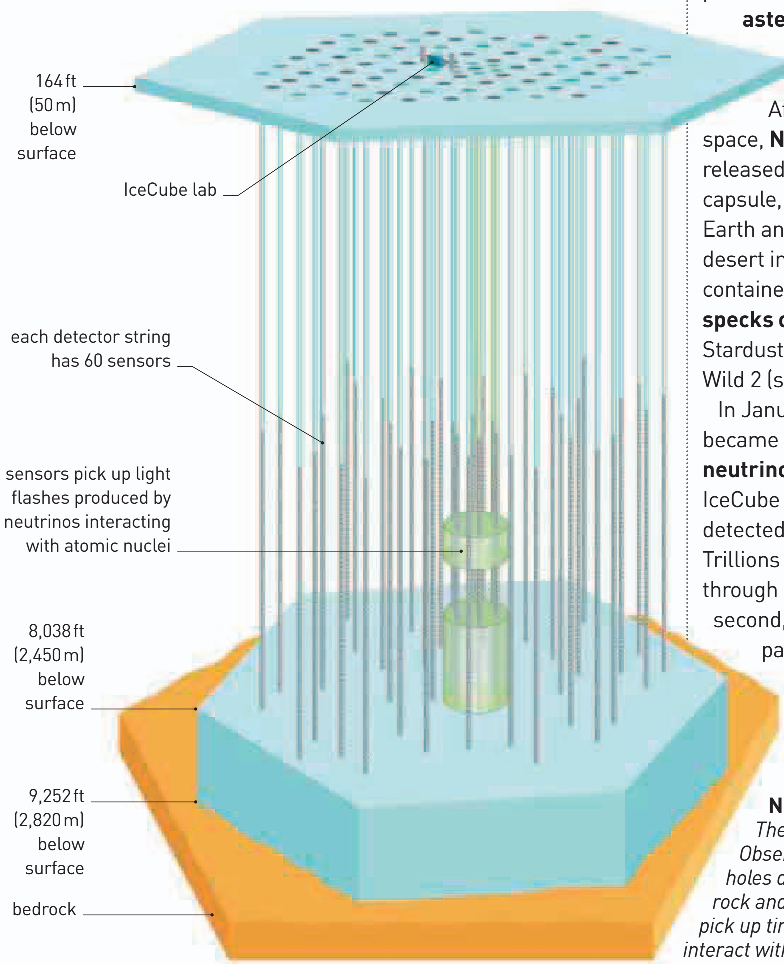


This carrot-shaped track in Stardust's aerogel collector panel was made by the tiny comet particle—the black speck to the right of this image—which entered the aerogel at a relative speed of several miles per second.

AT THE BEGINNING OF 2006, the Solar System officially had nine planets. The outermost, **Pluto**, stood out as being different from the others. For example, while all the other planets' orbits lie more or less in the same plane, Pluto's orbit is inclined at a fairly steep angle to it; Pluto's orbit is also very eccentric and passes inside Neptune's. When, in the 1970s, an object similar to Pluto was discovered, **Pluto's classification as a planet was in doubt**. In 2005, astronomers detected a rocky object outside the orbit of Neptune with a mass greater than Pluto's. After much deliberation, the International Astronomical Union decided in 2006 to designate Pluto as a dwarf planet, **one of many similar asteroid-like objects** in the Kuiper Belt (see 1949).

After seven years in space, **NASA's Stardust probe** released its sample return capsule, which parachuted to Earth and landed safely in the desert in Utah. The capsule contained about a **million specks of dust** collected from Stardust's encounter with Comet Wild 2 (see 1999).

In January, shortly after it became the **world's largest neutrino observatory**, the IceCube Neutrino Observatory detected its first neutrino. Trillions of neutrinos pass through the detector every second, but these elusive particles interact only very weakly with matter, making them very difficult to detect. Every so often, one will interact with the nucleus of an atom, resulting in a **tiny flash of light**. The observatory has strings of detectors in the rock and ice under the ground in Antarctica that pick up the tiny flashes of light. Neutrinos originating from high-energy phenomena, such as supernovas and mysterious gamma-ray bursts, provide astrophysicists with a **unique window on the Universe**. Construction of the observatory began in 2005, and was completed in 2010.



Neutrino observatory
The IceCube Neutrino Observatory consists of deep holes drilled into the Antarctic rock and ice. Strings of detectors pick up tiny flashes as neutrinos interact with atomic nuclei.

8 THE **NUMBER** OF **PLANETS** IN OUR **SOLAR SYSTEM** AFTER **AUGUST 24, 2006**

Just 10 years after the launch of DVDs (see 1995–96), this year saw the introduction of **two rival formats for video disks** designed to carry high definition video. HD DVD and BluRay can carry around **10 times as much data as DVDs**. Each had different electronics and entertainment companies bent on promoting it. Japanese company Toshiba was the main protagonist for HD DVDs, Sony for BluRay. Within just two years, **BluRay had won this format war**, largely because it garnered more support, but also partly as a result of the inclusion of a BluRay player in Sony's popular PlayStation 3 games console, which was released in November 2006.

In August, doctors in Australia administered for the first time a **vaccine for the human papilloma virus** (HPV). There are many different strains of HPV; several types are transmitted during sexual contact, and are the most common cause of genital warts and cervical cancer. Medical researchers suggested that the vaccine should be given to girls and boys routinely, in an effort to **reduce incidence of cervical and other cancers**. This idea was controversial, with some groups claiming it would encourage underage sex. Nevertheless, the vaccine is now routinely given in many countries.

In the journal *Nature*, a team of paleoanthropologists (who study hominid history through fossil evidence) described an

January 16 NASA's Stardust mission returns **samples taken from Comet Wild 2** to Earth

January 19 NASA's New Horizons **space probe sets off for Pluto**

January 29 IceCube Neutrino Observatory, near the South Pole, **detects its first neutrino**

April 18 The first film titles and players for **HD DVD format** are released

June Scientists **reprogram normal mouse cells** to produce stem cells—bypassing the need to use stem cells from embryos

June 20 The **first BluRay** titles and players are released

July 18 Release of Intel's Itanium 2 microprocessor, the first computer chip with **more than a billion transistors**

August 24 Pluto is officially **designated as a dwarf planet**

August 28 HPV vaccine is **approved in Australia** (elsewhere soon afterward), amid controversy

September 20 A skull and other bones from a **3.3 million-year-old human ancestor** are found in Ethiopia